Ojasva Mishra

Z Fellow | CMU SCS Researcher | SWE & AI/ML Systems | Quantitative Research Analyst | Robotics & Autonomy
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EDUCATION

University of Florida, Gainesville, FL
Bachelor of Science in Computer Engineering; dual major in Finance
Graduation Date: May 04, 2029
YC Startup School 2026 | Downingtown STEM Academy, IB Diploma | SAT 1590, National Merit Commended

EXPERIENCE

Drift November 2025 to Present
AI Platform Engineer (Founding Team) San Francisco, CA
- Built core infrastructure for an **Antler-backed robotics AI coding workspace used by 1,000+ users** to configure, run, and debug simulations across ROS2, Gazebo, MuJoCo, and Isaac Sim.
- Shipped **World Labs photo-to-3D scene generation**, a license-verified **129-robot catalog**, and autonomous OpenClaw + pexpect evals; traced a production Gazebo freeze to gz-transport UDP multicast.

Carnegie Mellon University, Xu Lab July 2025 to Present
Research Intern Pittsburgh, PA
- **First author on three robotics and control papers** with Prof. Min Xu and Dr. Xiaolong Wu covering actuator-aware control optimization, adversarial safety filtering, and secure feedback control.
- Validated algorithms on a **Franka Emika Panda**; built multimodal SLAM, path planning, and constrained control experiments from perception to actuation.

Twinly Incorporated December 2025 to Present
Co-Founder and CEO San Francisco, CA
- Built Twinly, which models a user's writing, speech, and computer behavior to create **digital clones** that handle work across hundreds of parallel instances; selected for **Z Fellows' 10-builder cohort**, with mentorship from founders of **Netflix, Eventbrite, and Tinder**.
- Shipped production **macOS, Windows, and iOS apps** serving **1,000+ active users, including paying customers**; built the Identity Runtime with app/browser context, persistent memory, and multimodal vision.

AlgoGators Investment Fund September 2026 to Present
Quantitative Research Analyst Gainesville, FL
- Selected as **1 of 2 first-year students** for the quantitative research team at a **~$500K AUM** student-run fund; build systematic signals, backtests, and trading strategies.
- Executed a **crack spread strategy with a 64% win rate across live trades**; adjusted positions as hurricane forecasts changed during an active storm.

EV Bots, Inc. April 2025 to February 2026
AI Robotics Engineer Intern Chester Springs, PA
- Built the company's **first functional autonomous EV-charging robot prototype** across mechanical design, electronics, perception and control, and closed-loop vehicle connection; designed an autonomous drone-charging dock. Work contributed to multiple provisional patents.

PteroLabs July 2025 to January 2026
Engineering Intern (UAV Simulation and Autonomy) Remote, United States
- Built a **mission-grade Unreal Engine 5 UAV simulator** with PX4, ArduPilot, and Betaflight SITL, 6-DOF flight dynamics, FPV, ML mission planning, and object detection; used by the **University of Pennsylvania** for active research.

iPipeline June 2025 to August 2025
Software Development Intern Exton, PA
- Shipped **enterprise software infrastructure**: automated test frameworks, AI agents, and developer tooling adopted across engineering teams for validation and regression workflows.

RESEARCH AND PUBLICATIONS

- **First author:** Constraint-Aware Discrete-Time PID Gain Optimization for Robotic Joint Control Under Actuator Saturation. **IEEE T-RO**.
- **First author:** Bounding Detector-Accepted Feedback Authority in Torque-Limited Robot Control. **IEEE ICRA 2027**.
- **First author:** SCIG++: Risk-Limiting Conformal Safety Filtering for Attack-Resilient Robotic Joint Control Under Saturation. **IEEE/RSJ IROS 2026** (video).
- Introduction to Robotics and Programming, a 120-page open-access textbook published through FusionBots and indexed on SSRN.

PROJECTS

Déjà View | SwampHacks XII, 1st AI/ML track and 1st DeepSpace track | Live | GitHub | Video | Devpost October 2026
- Turns photos or video of a space, real or in-game (Fortnite's Tilted Towers), into a 3D world you walk into on Meta Quest (WebXR), in a browser, or in Roblox: COLMAP camera solve (163 cameras, ~1 px reprojection error), then a ~1M-Gaussian splat trained with gsplat MCMC on UF HiPerGator (30k steps in 11 min).
- Checks every Gaussian against the cameras that saw it and tints the guessed parts; fills gaps from 2,304 probe views with Qwen2.5-VL, Qwen-Image-Edit and Depth Anything V2, gated at 45 dB PSNR; fits people from video with per-frame depth and Multi-HMR (mask IoU 72% to ~85%); compiles each world to a life-size Roblox map (3.57 studs/m) with 240 Hz ping pong physics shared by VR and Roblox players.

Courtside | Gator Quant Hacks 2026, Systematic Trading track | Paper | GitHub | Video October 2026
- Built a computer-vision ball tracker that calls a point (out or into the net) before the ball lands: on held-out table tennis, early calls come a median 162 ms before the bounce, none wrong, at 119.9 fps; on broadcast tennis, out calls were right 5 of 5 at the bounce.
- Tested what that speed is worth on 13,084 Polymarket ATP/WTA matches: wallets trading within 3 s of a point earned after fees in all 11 months, as a pre-registered hypothesis predicted out of sample; priced the first second of feed delay at $42 to $74 a day. Paper trading only.

COMPETITIONS AND LEADERSHIP

- **American Rocketry Club, Founder and President:** grew to 60+ members; led payload CAD, propulsion analysis, systems integration, and launches.
- **VEX Robotics, 3x World Championship Competitor:** captain, lead programmer, and driver; 7 seasons of C++ and Python autonomy, PID tuning, CAD, and sensors.
- **Technology Student Association, 2x National Finalist:** 8 national events; captain for 6. Projects: SAFMS AI/IoT, drone challenge, engineering design, systems control, robotics, VR visualization, video game design.
- **Hackathons and builds:** SwampHacks XII, **1st** AI/ML track and **1st** DeepSpace track (Déjà View); Gator Quant Hacks 2026, Systematic Trading track (Courtside); YC Transpose, **1st** (HIVE); Startup in a Weekend Philly, **2nd** (GlassBoxAI); NexHacks (Kinetix); Eyesight; PreventaAI; QuickCitePro; FusionBots.
- **Other leadership:** Co-founder of FusionBots, a robotics nonprofit serving 200+ students; CTO of Youth Publications and Socioeconomic Forum, leading AI literacy programs across 15+ countries.

TECHNICAL SKILLS

Languages and software: Python, C++, Swift, C#, JavaScript, TypeScript, SQL, Linux, Git, gRPC, React, Next.js, Tauri, Electron, Supabase, Firebase, Stripe
AI and ML: PyTorch, LLM agents, multimodal and VLM systems, computer vision, reinforcement learning, Claude Code, OpenAI Codex, Cursor, CoreML, MediaPipe
Quantitative research: Backtesting, systematic trading strategies, crack spread analysis, market data analysis, forecast-driven trading
Robotics and aerospace: ROS2, SLAM, path planning, PID control, Gazebo, MuJoCo, Isaac Sim, PX4, ArduPilot, Betaflight, Unreal Engine 5
Hardware and product: Arduino, VEX, Fusion 360, Onshape, 3D printing, CNC machining, woodworking, metal cutting, band saw, miter saw, SwiftUI